RESEARCH PROJECT REPORT

FINAL REPORT

MAKING EDUCATION MORE ACCESSIBLE & RELIABLE : AI–DRIVEN FACILITATOR–BASED TEACHING

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ABSTRACT

Facilitator-based delivery has the potential to extend access to education by decoupling the design of the lesson from the actual delivery of the lesson. It does so at the expense of relying on facilitators (who are typically less experienced than qualified teachers) to identify which students are having difficulty. This report looks at the facilitator/student side of a two-part project: a proof-of-concept facilitator decision-support tool for Year 3–6 reading comprehension. The tool is intended to be a feature of the partner's lesson delivery platform (StudyBuddy). Before or after each session, students complete a short reading check (a passage with two literal and two inferential questions) and judge for themselves whether they got each question right. Over time, repeated checks build up a body of evidence for each individual student. The system records accuracy by question type, self-judgments, hint timing, revisions and timing. Rules derived from reading comprehension research take these signals and produce a status ("needs support" or "worth a check-in") with the evidence behind it.

With no real classroom data available, the rules were evaluated on 200 simulated classrooms of 30 learners built from five reader profiles and compared with an accuracy-only threshold and two Random Forest models. Results are reported for the more cautious timing-neutral scenario, in which the poor comprehenders do not give themselves away by answering quickly. On a single passage, the refined rules achieved 64.0% detection of at-risk learners (95% CI 61.0-66.9%) compared to 59.3% for accuracy alone, with fewer false positives (8.8% vs 10.8%). They also detected a greater proportion of the difficult-to-spot poor comprehenders (53.7% vs 46.4%). A Random Forest trained on the rule-generated labels replicated the earlier rules (48.8% recall). A Random Forest trained on the true simulated labels reached 62.6% on one passage and 84.2% on three, which indicates what real outcome labels could add. Pooling the three passages increased the rules' precision to 94.3%, though a threshold granularity issue reduced detection of poor comprehenders to 38.8%.

The results show that evidence-based rules are not only feasible but also slightly better than a score in simulation. The persona parameters are all assumptions, though, so validation with actual learners and facilitators is the crucial next step.

SIGNED STATEMENT OF CONTRIBUTION

Student

I hereby declare that:

1. This report is the result of the final year project work carried out by my project partner (see cover page) and I under the guidance of our supervisor (see cover page) in the 2026 academic year at the Department of Electrical, Computer, and Software Engineering, Faculty of Engineering and Design, University of Auckland.

2. This report is not the outcome of work done previously.

3. This report is not the outcome of work done in collaboration, except that with a potential project sponsor (if any) as stated in the text.

4. This report is not the same as any report, thesis, conference article or journal paper, or any other publication or unpublished work in any format.

In the case of a continuing project, please state clearly what has been developed during the project and what was available from previous year(s):

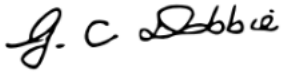
N/A

Signature: 

Date: 17/10/2026

Supervisor

I confirm that the project work undertaken by this student in the 2026 academic year is not part of a continuing project, components of which have been completed previously. Comments, if any:

Signature: 

Date: 18/10/26

Table of Contents

Acknowledgements vi
Glossary of Terms vii
Abbreviations viii
1. Introduction 1
1.1. Background and Motivation 1
1.2. Project Structure: Two Complementary Halves 1
1.3. How the Tool Is Used 2
1.4. Problem Statement 3
1.5. Research Aims and Questions 3
1.6. Objectives 3
1.7. Scope 4
1.8. Report Structure 4
2. Literature Review 4
2.1. Overview of Artificial Intelligence in Modern Education 4
2.2. Applications and Benefits of AI in Younger Education (K–12) 5
2.3. AI in Self-Directed Learning (SDL) 5
2.4. Limitations of AI for Younger Learners (K–12) 6
2.5. Implications for AI-Supported Learning System Design 7
2.6. Identified Gaps in AI-Supported Education 7
2.7. Reading Comprehension in Years 3–6 8
2.8. Learning Analytics and Evaluation with Simulated Learners 9
3. Methodology 10
3.1. Design Approach 10
3.2. Activity Design Grounded in Reading-Comprehension Research 11
3.3. Signals and Data Model 12
3.4. Recommendation Rules 12
3.5. Evaluation Method: Simulated Classrooms 13
3.6. Compared Systems 14
3.7. Metrics and Robustness Checks 14
3.8. Implementation and System Features 14
4. Results 15
4.1. Overall Comparison (RQ1) 16
4.2. Detecting Poor Comprehenders (RQ2) 16
4.3. More Data per Learner 17
4.4. Machine Learning Comparison (RQ3) 18
4.5. Robustness and Sensitivity 18

4.6. Workload 19
4.7. End-to-End Traceability Walkthrough 19
5. Discussion 20
5.1. Answering the Research Questions 20
5.2. Relation to Literature 20
5.3. Contribution 21
5.4. Design Trade-offs and Ethical Considerations 21
5.5. Limitations and Threats to Validity 22
6. Conclusions and Future Work 22
6.1. Conclusions 22
6.2. Future Work 23
References 24
Appendix A Simulation Parameters and Rule Thresholds 28
A.1. Persona Parameters 28
A.2. Rule Thresholds 28
Appendix B Reproducing the Results 30

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Glossary of Terms

Term	Definition
Answer-key model	A Random Forest trained on the true simulated labels; a ceiling, not deployable
Bias score	Mean of judged correctness minus actual correctness; positive means overconfident
Decoding-limited reader	A reader whose word reading is slow or inaccurate, limiting comprehension
Evidence-linked	A recommendation that shows the raw data that produced it
Facilitator	A person who delivers a lesson designed by someone else, often without full teacher training
False-alarm rate	Share of not-at-risk learners that a system flags
Indicator	One rule's output: a severity, the evidence that triggered it and an interpretation
Inferential gap	Literal accuracy minus inferential accuracy for a learner
Inferential question	A question that requires combining passage information with reasoning or background knowledge
Literal question	A question answered by information stated directly in the passage
Overconfidence on errors	The share of a learner's wrong answers that they judged correct
Persona	A simulated reader profile with assumed behaviour distributions
Poor comprehender	A reader who decodes fluently but has weak comprehension, especially inference
Postdictive judgement	A learner's judgement, made just after answering, of whether their answer was correct
Pre-attempt hint	A hint requested before the learner selected any answer
Precision	Share of flagged learners who are actually at risk
Pseudo-label	A training label produced by rules rather than by observing real outcomes
Reading check	The short (five to ten minute) activity learners complete before or after each session: a passage, four questions and a self-judgement after each answer
Recall	Share of at-risk learners that a system flags
Repair	Rereading the passage or revising an answer after a hint
Simulated learner	A computer-generated learner used to evaluate an educational system
StudyBuddy	The partner project's lesson creation and delivery platform (M. Manglani); not the tool presented in this report
Timing-neutral	The headline scenario, in which poor comprehenders are given typical timing and help-seeking behaviour
Year 3–6	New Zealand school years for children aged about 7–11
α (alpha)	How strongly persona parameters differ from typical; 1 = as defined, 0.5 = half the difference

Abbreviations

AI	Artificial Intelligence
AIED	Artificial Intelligence in Education
CI	Confidence Interval
CSV	Comma-Separated Values
K–12	Kindergarten to Year 12 (school education)
ML	Machine Learning
RQ	Research Question
SDL	Self-Directed Learning

1. Introduction

1.1. Background and Motivation

Access to reliable, good-quality education remains a global issue. It is estimated that 44 million additional teachers will be required worldwide by 2030 [1] and 70% of 10-year-olds in low- and middle-income countries cannot read and comprehend a simple text [2]. Technology has widened access to learning materials but not to effective learning. Less than 15% of learners actually complete an unsupervised online course [3], and as Section 2 will show, AI tools tend to assume a level of self-direction that young learners simply do not yet possess. School practitioners consulted during the project raised the same issue, pointing to variation in teaching quality from class to class, the loss of continuity when a teacher leaves, and the reality that classroom management is as important as subject knowledge.

This project (Part IV Project #133) explores decoupled delivery: separating who designs a lesson from who delivers it. Subject experts design lessons once. Any approved facilitator, who need not be a qualified teacher or subject expert, then delivers them with structured, step-by-step support. AI supports this model rather than replacing the human in the room, for example, by drafting materials and by helping facilitators see which learners need attention.

1.2. Project Structure: Two Complementary Halves

Decoupled delivery raises two distinct questions, so after a shared start, the project was divided into two halves, each producing its own artefact with its own literature review, research question, methodology, evaluation and report (Table 1.1). The admin–facilitator side, by M. Manglani, produced StudyBuddy, a lesson creation and delivery platform. The facilitator–student side, presented in this report, produced a facilitator decision-support tool: a proof-of-concept design that turns learners' interaction data into evidence-linked recommendations. The shared work covered the problem definition, the decoupled-delivery concept and initial background research.

Table 1.1: The two halves of Project #133.

	Admin–facilitator side: StudyBuddy (partner report, M. Manglani)	**Facilitator–student side: decision-support tool (this report)**
Focus	Designing and delivering the lesson	Seeing which learners need help, and why
Research question	Can any approved facilitator, regardless of subject expertise, deliver an expert-designed lesson?	Can a software system convert learner interaction data into evidence-linked recommendations for facilitator decision-making?
Artefact	StudyBuddy: admins author lessons from blocks, facilitators deliver them step by step, AI drafts worksheets for admin approval	Proof-of-concept tool: reading activity, evidence-linked rule engine and facilitator dashboard
Evaluation	Practitioner feedback and usability heuristics	200 simulated classrooms of 30 learners

The two sides come together at the facilitator. StudyBuddy ensures that a facilitator can deliver the lesson as designed. The decision-support tool addresses what the delivery of the lesson alone cannot achieve: a facilitator without any teaching experience is simply not in a position to spot which learners are having difficulties, particularly when that difficulty is not apparent.

This report is solely about the decision-support tool. It makes no evaluation of StudyBuddy's lesson authoring or delivery capabilities, and the two prototypes have not yet been integrated; Section 1.3 describes how they are intended to work together.

The tool focuses on reading comprehension in Years 3–6, when many children begin to fall behind [4]. Because no real classroom data was available, it was evaluated on simulated classrooms against an accuracy-only threshold and a machine-learning model.

1.3. How the Tool Is Used

The decision-support tool is a short, low-stakes reading check that learners complete before or after each session. It takes 5 to 10 minutes, but a single check gives only four answers per learner, which is too few to judge reliably (Section 4.3). Its value comes from repeating it: over time the checks build a baseline for each learner, show trends across sessions and let each learner be compared with their own earlier performance rather than a fixed standard. The facilitator sees this in two views, an individual learner profile and a class view of common difficulties. The check is not a test and is not marked; it helps the facilitator decide who to support and how, without replacing their judgement.

The tool is designed to work as a feature of StudyBuddy rather than as a separate product. In StudyBuddy, an admin builds a lesson from blocks and a facilitator delivers it step by step on a shared screen. Table 1.2 shows the intended workflow.

Table 1.2: Intended workflow when the decision-support tool is used with StudyBuddy.

Step	Who	Where	What happens
1	Admin	StudyBuddy lesson editor	Attaches a short reading check (a passage with two literal and two inferential questions) to each lesson.
2	Learners	Reading check (this tool), before the session	Read the passage on their own devices, answer the questions and judge each answer; the tool records the interaction signals (Section 3.3).
3	Facilitator	Support view (this tool)	Before teaching, sees who needs support today, why, and a suggested next step, plus class-wide patterns to address in the lesson.
4	Facilitator	StudyBuddy delivery	Delivers the lesson step by step, giving extra attention to the learners and topics flagged.
5	Learners	Reading check (this tool), after the session (optional)	A second short check shows whether difficulties persisted; across sessions, the checks form each learner’s baseline and trend.
6	Admin	StudyBuddy	Receives class-pattern alerts (many learners missing the same question) as feedback for revising the passage or lesson.

StudyBuddy makes sure lessons are delivered as designed, and the decision-support tool shows facilitators how learners are doing, session by session. Step 6 closes a feedback loop that decoupling would otherwise break: lesson designers are not in the room, so class-level evidence is how they learn whether their materials work. The proof of concept implements Steps 2 and 3 and the class-pattern alerts. Each check is stored as a separate attempt, the dashboard shows each learner's attempt history, and the comprehension-pattern rule pools the last three checks. Before-and-after comparison, trend views and integration with StudyBuddy's database are designed but not yet built (Section 6.2).

1.4. Problem Statement

In spite of advances in the field of educational technology, there are still massive problems with delivering consistent, high-quality education at scale. Education still relies heavily on the teacher for both lesson design and delivery, creating a structural bottleneck and leading to wide variation in learning quality, particularly in the most disadvantaged or under-resourced areas. Online and AI-supported platforms provide flexibility, but they tend to assume a level of learner autonomy that is simply not appropriate for young children, who need lesson structure, supervision, and the interactivity a teacher provides. Furthermore, access to AI-enhanced content does not guarantee consistent or effective instructional lesson delivery; variability in teaching approaches, the absence of lesson structures and limited contextual adaptability continue to produce inequitable learning outcomes. These compounding limitations point to a clear gap in current provision: the lack of a scalable system which separates expert lesson design from classroom delivery and provides the required structure of educational support throughout the learning journey.

On the facilitator–student side, the gap takes a specific form. Once lessons are delivered by facilitators rather than teachers, nobody in the room may have the expertise to notice which learners are struggling, or why.

1.5. Research Aims and Questions

The literature in Section 2 demonstrates that AI tools for K–12 learners assume a level of self-direction that many young learners simply do not have. Facilitator-led delivery in that context is contingent on the facilitator being able to spot who is struggling. This project therefore investigates whether a lightweight system can take the interaction data from a brief reading task and turn it into evidence-based recommendations that would assist a less experienced facilitator in determining which learners require some form of support. The core research question is:

Can a software system convert learner interaction data into evidence-linked recommendations for facilitator decision-making?

Applied to Year 3–6 reading comprehension, three sub-questions follow from it:

1. **RQ1.** Does a rule engine grounded in reading-comprehension research (rules v2, Section 3.4) identify the learners who require support more effectively than a simple score cut-off, and what is the cost in false positives?
2. **RQ2.** Can the rule engine identify *poor comprehenders* (fluent readers with weak inference skills that teachers are likely to miss) via the literal/inferential gap and overconfidence on their errors?
3. **RQ3.** Does a machine-learning model (Random Forest) improve on the rules, given that no real classroom outcome labels are available?

The questions are answered on simulated classrooms, not real learners (Section 3.5), so the answers describe how the design behaves under stated assumptions rather than how it will perform in a school.

1.6. Objectives

For the facilitator–student side, the project aims to:

- Investigate challenges related to access, quality and teacher dependency in education.

- Analyse current AI-based and traditional learning systems and their limitations for younger learners.
- Identify, from reading-comprehension research, interaction signals that indicate a learner needs support.
- Design and build a reading activity, rule engine and facilitator dashboard that turn those signals into evidence-linked recommendations.
- Evaluate the rules against a score baseline and machine-learning models on simulated classrooms.
- Assess the limitations of the approach and the evidence needed before real-world use.

Objectives three and four produce the rule engine examined in RQ1 and RQ2, objective five answers RQ1–RQ3, and objective six is addressed in Section 5.

1.7. Scope

The scope is a proof of concept of facilitator decision support for Year 3–6 reading comprehension. It covers the student reading activity, the recommendation rules, the facilitator dashboard and a simulated evaluation. It excludes lesson authoring and delivery, which StudyBuddy and the partner report cover, and any trial with real learners or facilitators.

1.8. Report Structure

Section 2 reviews the literature on AI in K–12 education, reading comprehension and evaluation with simulated learners. Section 3 describes the design, the rules and the evaluation method, and Section 4 presents the results. Section 5 discusses their meaning, trade-offs and limitations, and Section 6 concludes with directions for future work. Appendix A lists the full simulation parameters and rule thresholds, and Appendix B explains how to reproduce the results.

2. Literature Review

2.1. Overview of Artificial Intelligence in Modern Education

The growth of AI in education is closely tied to the growth of online learning. As more teaching moved online, AI tools were built in to support content delivery, interaction and assessment [5], [6]. This is part of a wider shift away from classroom-based teaching towards flexible, technology-enabled learning, where students can work on digital platforms at their own pace without constant teacher oversight [7]. The teacher's role has shifted with it, from direct instruction towards facilitation and support [8], with more emphasis on student-centred, personalised learning.

Figure 2.1 shows that AI in education can be thought of in three stages: function, utilisation and impact. The framework describes how AI moves from a technical capability to an actual effect on learning outcomes [9], and gives a starting point for looking at how AI is used in classrooms.

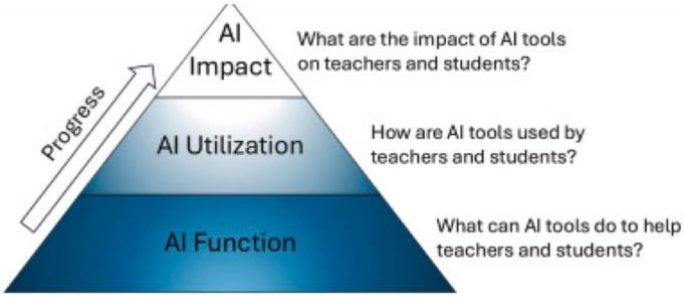


Figure 2.1: Progressive stages of AI application in education (adapted from [9]).

However, the move to AI platforms has not benefited all learners equally. Access has improved, but the quality of the experience varies widely: some learners cope well on their own and others do not [10]. AI has also been adopted faster than good implementation frameworks have been developed, so it is applied unevenly across schools [9]. This tension between flexibility and consistency shows why it matters to understand how AI supports learning, especially for younger learners who depend on structure and guidance.

2.2. Applications and Benefits of AI in Younger Education (K–12)

AI is used in education to personalise learning, automate tasks and support teaching. The best-known example is adaptive learning systems, which tailor content to each student and let them work at their own pace [9], [11]. In a study of 269 primary school pupils, AI-based personalised learning significantly improved performance [11]. AI can also give instant feedback, so students see where they went wrong and correct it straight away [12]. This is a real advantage over traditional teaching, where feedback often comes later, especially in practice-heavy subjects such as maths and languages.

AI has also improved accessibility and efficiency. Students can reach learning materials at any time and from anywhere, which matters most in remote or under-resourced areas with few traditional resources [13]. AI can adapt content to different learning styles and abilities, and automate administrative work such as marking, content creation and data analysis [14]. This frees teachers to spend more time with students and helps learning programmes scale.

Even so, the evidence on AI's effectiveness is mixed. An umbrella review of 102 studies found that AI increased engagement and personalisation, but how well it worked depended on the quality of implementation and on how students interacted with it [9]. Some studies found strong gains in learning outcomes; others found very little. AI on its own does not guarantee better learning. This is especially true for self-directed learning, where AI's effectiveness depends heavily on the learner's own skills and habits.

2.3. AI in Self-Directed Learning (SDL)

Self-directed learning (SDL) is an educational approach in which students assume responsibility for planning, monitoring and evaluating their own learning. Research indicates that AI tools can facilitate SDL by offering personalised guidance, feedback and adaptive learning pathways, thereby encouraging students to take a more active role and enhancing motivation and engagement [15], [16]. A meta-analysis of 32 studies with 3,029 participants found that AI-based interventions significantly improved both self-regulated and self-directed learning [16]. Similarly, a study involving 74 students over three weeks reported improvements in engagement, motivation and learning outcomes [15].

As illustrated in Figure 2.2, AI interventions support different dimensions of self-regulated learning, including cognitive, motivational and behavioural processes, which contribute to the development of self-directed learning [16].

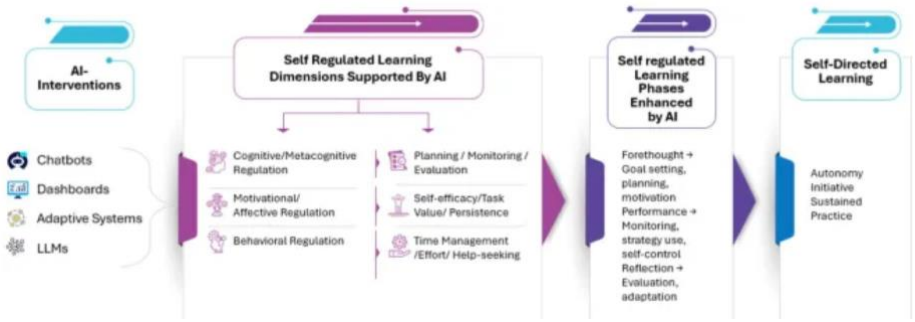


Figure 2.2: AI–regulation–autonomy pathway in learning (adapted from [16]).

This demonstrates AI's function as a scaffold that assists learners in planning, monitoring and evaluating their progress. However, its effectiveness is contingent upon the learner's ability to engage independently with these processes. Research consistently shows that students need strong self-regulation skills, such as time management, discipline and progress monitoring, to use AI tools effectively [17]. Earlier studies also found that students in online environments often struggled with independent learning [18], [19]. Without these foundational skills, students not only failed to benefit from AI's features but also risked developing over-reliance, which is associated with diminished critical thinking and problem-solving abilities [5].

Taken together, these findings indicate that AI is most effective when learners demonstrate strong self-directed learning skills, a trait less frequently observed among younger students in K–12 settings. Consequently, there are significant concerns about the suitability of AI-driven approaches for this age group, which will be examined in the following section.

2.4. Limitations of AI for Younger Learners (K–12)

Despite its potential, AI has several limitations for younger learners. The first is the assumption of self-directed learning: younger learners often struggle to learn on their own and, as a result, tend to be less engaged and achieve less [18]. Reports have also found that students often rely on AI to complete their work without engaging with the material, in some cases bypassing essential learning processes altogether [20]. A review of 22 studies identified further risks, including cognitive dependency, reduced creativity and harm to wellbeing [21], raising concerns that AI may replace learning rather than support it.

AI also cannot provide the structured guidance and meaningful human interaction that young learners need [22]. Unlike teachers, it cannot properly respond to learners' emotional and behavioural needs, which may reduce engagement and motivation, especially in younger children. Other reported risks include reduced confidence, social isolation and dependence on technology [21], which matter most for learners who are still developing core cognitive and social skills. Wider issues of data protection, ethics and unequal access to technology add to the uneven way AI is implemented and experienced across different contexts [9].

In summary, these findings show that although AI offers potential benefits in K–12 education, it does not fully meet the developmental and pedagogical needs of younger learners.

2.5. Implications for AI-Supported Learning System Design

The literature suggests that integrating AI well requires careful system design that accounts for both its benefits and its limits. AI is consistently presented as a tool that supports teachers rather than replacing them, and that helps make learning more structured and meaningful [5].

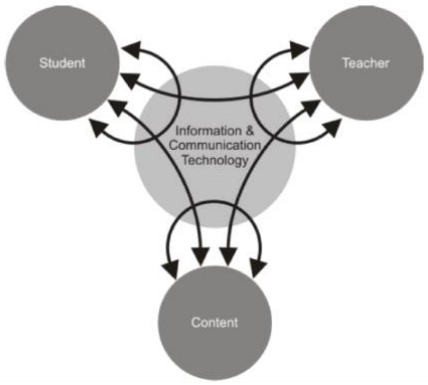


Figure 2.3: Interaction between students, teachers and learning content in AI-supported learning environments (adapted from [11]).

Figure 2.3 shows how AI can connect students, teachers and learning content in a more adaptive learning environment [11]. This supports the idea that AI works best inside a structure. Guided learning environments, where students move through clear steps with consistent support, help keep them engaged and stop them becoming confused or disconnected [9]. In this approach the facilitator stays at the centre, guiding students, monitoring progress and giving extra support when needed, which combines AI's scale with the value of human interaction [14]. AI should strengthen core learning rather than replace it [12], for example in content creation, feedback and practice, while the facilitator remains responsible for learning quality. Centralised content and structured lesson design can then give every student the same quality of learning, wherever they are and whoever their facilitator is.

Overall, the literature points towards embedding AI in structured, facilitator-led learning: AI can offer scale and flexibility, but not the consistency and human connection that learning needs.

2.6. Identified Gaps in AI-Supported Education

Building on the sections above, the literature points to several lasting gaps in how AI is used in K–12 education. AI's potential to improve personalisation, access and efficiency is widely recognised, but the systems in use often do not suit young learners' needs or the practical realities of delivery at scale.

Personalisation and the Delivery Gap. Research consistently shows that personalised learning improves engagement, achievement and motivation by tailoring content to each learner [11], [12], and AI does this well through adaptive content and real-time feedback. However, current systems focus heavily on personalising content and much less on how learning is structured and delivered. A student may receive tailored material while the learning experience around it stays inconsistent and dependent on context. Effective AI-supported learning needs both: personalised content and well-structured delivery.

Dependence on Teacher Expertise and Scalability Constraints. Current models depend heavily on teacher expertise for both designing and delivering learning. AI can help with content and feedback, but structuring and facilitating learning still falls to the teacher [9]. This

creates a scalability problem wherever skilled teachers are scarce or unreliable. In current systems, lesson design and classroom delivery are not separated enough, which limits the reach and consistency of AI-supported education, and the literature does not yet offer a clear solution.

Lack of Structured Guidance in AI-Supported Learning. Many AI tools assume learners can direct their own learning. As Sections 2.3 and 2.4 showed, younger learners often lack the self-regulation skills to use AI independently and effectively [17], [18]. Without structure, they can become confused or disengaged, or treat AI as a 'black box' that produces answers, which is especially harmful while foundational skills are still developing. Flexibility is being prioritised over structure, which undermines AI for the learners who need the most support.

Accessibility and Equity Gap. AI has improved access to learning materials through flexible, on-demand delivery [13], but access to good-quality education is still uneven. Teacher availability, school resources and location still shape outcomes in ways AI cannot fix on its own [9]. Current implementations improve access to content rather than ensuring good learning experiences in every setting, and system design needs to address that difference directly.

Limited Adaptability Across Contexts. AI systems can personalise content for individual learners, but many do not adapt well across contexts. Differences in curriculum, age, subject and culture are not always accommodated, which limits how well systems transfer between settings [16]. Most studies also test AI tools in a single institution or region, so their results are hard to generalise. Systems need to support flexible design at the platform level, so that both content and delivery can be adapted to different contexts.

Synthesis. Overall, the literature shows a consistent pattern. Current AI-based systems focus on personalising content rather than delivery, depend on teachers in ways that limit scale, assume more learner independence than many young learners have, improve access to content without ensuring learning, and do not adapt well across contexts. AI's potential in education is therefore not yet realised, particularly for K–12 learners in non-traditional or under-resourced settings. Closing these gaps is the basis for the system design in this study.

2.7. Reading Comprehension in Years 3–6

Sections 2.1–2.6 treat AI in K–12 broadly. To design a tool that helps facilitators notice struggling learners, the project needed a narrower question: what does struggle look like in a specific skill? Reading comprehension was chosen because it underpins learning in every other subject, and because its difficulties are well characterised in the research literature.

Two Components of Reading. The Simple View of Reading [23] describes comprehension as the product of decoding (recognising words) and language comprehension (understanding what they mean together). A weakness in either component limits reading, but the two weaknesses look different in class. A learner with poor decoding reads slowly and effortfully, which is visible to any observer. A learner with poor language comprehension may read aloud fluently while understanding little, which is not.

The Year 3–6 Window. Chall's stage model describes a shift around Year 4 from "learning to read" to "reading to learn" [4]. Texts become longer and less familiar, and the demand for inference grows. Many learners who coped with early reading begin to fall behind at this point, the "fourth-grade slump". It is also the stage at which comprehension difficulty first becomes separable from decoding difficulty, which makes Years 3–6 the natural target for a tool that distinguishes them.

Literal and Inferential Comprehension. Literal questions ask for information stated in the text, while inferential questions require the reader to combine information or bring background knowledge. A longitudinal study of listening comprehension in 100 bilingual 5–6-year-olds found that literal comprehension and global inference improved over a year while local inference did not [24]. That study measured listening rather than reading, but it suggests the two kinds of comprehension do not develop in step. A single accuracy score blends them, so a learner who answers every literal question and misses every inferential one looks like a 50% reader rather than a reader with a specific, teachable gap.

Poor Comprehenders. Nation et al. [25] described children who decode at age-appropriate levels but have weak oral language and inference skills. They called them a hidden group, because their fluent reading masks the difficulty. Kelso et al. [26] found this profile in 24 of 218 children (11%). Teachers are not well placed to spot them. Schmitterer and Brod [27] found that teachers' judgements of reading difficulty aligned more closely with spelling results than with the reading skills in question, so a fluent reader with poor comprehension is likely to be judged as fine.

Metacognitive Monitoring. Good readers notice when they do not understand and act on it: they reread, ask for help or revise. Comprehension monitoring is therefore usually described as detection followed by repair [28]. Children with comprehension difficulties monitor less accurately. In a study of Year 3–5 readers, those with comprehension difficulties were worse at judging, immediately after answering, whether their answers were correct [29]. Overconfidence on wrong answers, rather than low confidence in general, is the signal most closely tied to comprehension difficulty.

Help-Seeking and Answer Changes. Not every visible sign of effort indicates struggle. Hints requested before any attempt are a stronger sign of difficulty than hints in general [30], and, at least among adult test-takers, most answer changes are from incorrect to correct [31]. Counting raw hints or revisions would therefore flag careful, self-correcting learners as struggling.

Together these findings identify a gap that general AI-in-education tools do not address. The learners least visible to a non-specialist facilitator, poor comprehenders, are characterised by signals a short digital activity can capture: the literal–inferential gap and overconfidence on errors. A facilitator tool that surfaces those signals would target the teacher-expertise gap identified in Section 2.6 at the point where it matters most.

2.8. Learning Analytics and Evaluation with Simulated Learners

Learning analytics dashboards present learner data to teachers so they can act on it [32]. Research on teacher dashboards repeatedly finds that teachers use them only when they understand what a number means and what to do about it [33]. This supports the project's evidence-linked design, in which each flag shows the raw counts behind it and a suggested next step rather than an opaque risk score.

Evaluating such a tool normally requires real learner data with known outcomes. When that is unavailable, AI in Education (AIED) has a long tradition of evaluating with simulated learners. Self [34] described a computational student model as a "wind tunnel" for testing tutoring designs before deployment. Early examples include Sierra [35] and Cascade [36]. Greer and Mark [37] list simulated learners among the standard methods for evaluating intelligent tutoring systems. The approach continues in SimStudent [38] and the TutorGym testbed [39].

Simulation offers known ground truth, control over each factor and the ability to run many classrooms.

Its weakness is validity: a simulation only shows how a system behaves on the learners its designer imagined. Käser and Alexandron [40] reviewed simulated-learner studies and proposed a "Turing-like test" of whether simulated learners can be distinguished from real ones. Few studies apply it. This project uses simulated learners for the same reasons as this literature, and treats the absence of such validation as its main limitation (Section 5.5).

3. Methodology

3.1. Design Approach

The project was a research-informed design. The requirements were established from the literature (Section 2), a prototype was built to meet those requirements and a simulated evaluation was then used to test the prototype against the simpler alternatives. Figure 3.1 shows the tool's pipeline. The learner completes a short reading task, the system derives a set of signals from their interaction with the task, rules are applied to the signals to generate flags with associated evidence and the facilitator dashboard presents a status, a recommended next step and class wide patterns.

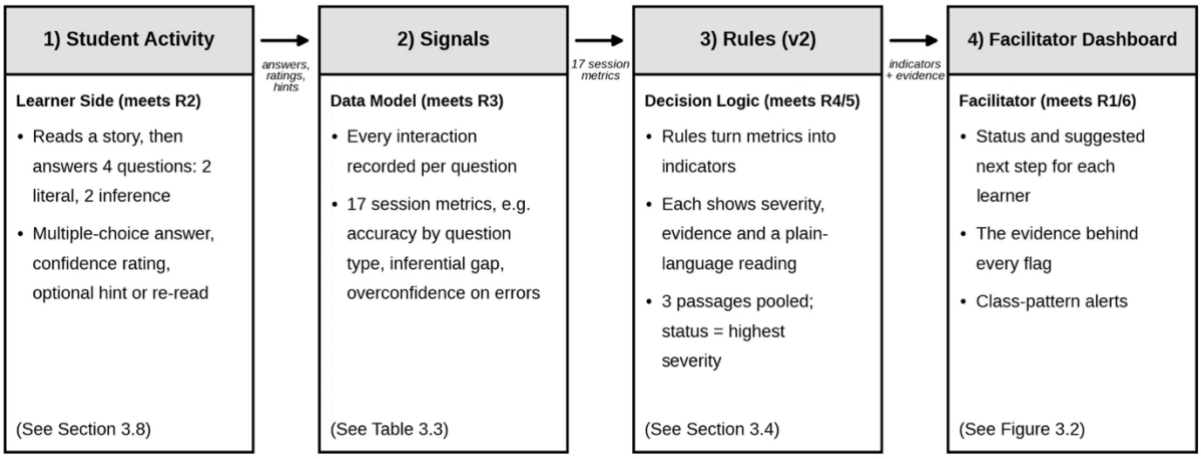


Figure 3.1: System pipeline from student activity to facilitator dashboard.

Two principles guided the design. First, every recommendation had to be evidence-linked: the facilitator sees the raw counts behind a flag, not just a label, because a less-experienced facilitator needs to know why a learner is flagged before acting. Second, the system should support the facilitator's judgement rather than replace it, in line with the hybrid model in Section 2.5. Transparent rules were therefore chosen as the decision mechanism, and a machine-learning model was evaluated alongside them (Section 3.6) to see whether it added value.

Design Requirements. Table 3.1 maps each requirement to the gap in the literature that it addresses and to the feature that fulfils it. The table also defines the scope: requirements which are raised in the literature but which this prototype does not meet are listed as out of scope rather than as omitted.

Table 3.1: Requirements derived from the literature.

Gap (section)	Requirement	Implemented as
Dependence on teacher expertise (2.6, 2.7)	R1. Help a non-specialist facilitator decide who needs support and what to do next	Status per learner, plain-language interpretation and suggested next step
Lack of structured guidance (2.4, 2.6)	R2. Keep the learner's task short, linear and guided; no self-directed navigation	Fixed sequence: instructions, passage, four questions, reflection
Personalisation without delivery (2.6)	R3. Point the facilitator to the type of difficulty, not just its size	Separate literal and inferential accuracy; distinct indicators for processing, accuracy, miscalibration and the comprehension pattern
Teacher judgement misses poor comprehenders (2.7)	R4. Detect fluent readers with weak inference and overconfidence	Comprehension-pattern rule combining the inferential gap with overconfidence on errors
Trust and transparency in analytics (2.8)	R5. Every flag must show its evidence	Each indicator stores the raw counts that triggered it
Variable delivery quality (2.6)	R6. Distinguish learner problems from material problems	Class-pattern alerts when many learners miss the same question
Adaptability across contexts (2.6)	Out of scope: one subject, one age band, three passages	Future work (Section 6.2)

3.2. Activity Design Grounded in Reading-Comprehension Research

The activity is aimed at Years 3 to 6 (the period when reading moves from decoding familiar text to reading to understand more complex text and when the difficulty of comprehension begins to be separable from the difficulty of decoding) [4]. The prototype consists of three short narrative passages, each followed by four multiple choice questions. Each design decision is underpinned by a finding from the literature (see Table 3.2).

Table 3.2: Activity design decisions and their basis in the literature.

Design decision	Rationale	Source
2 literal + 2 inferential questions per passage, scored separately	Literal and inferential comprehension develop on different trajectories; a blended score hides an inference gap	[24]
Postdictive judgement after each answer ("Do you think you got that right?": yes / not sure / no)	Children with comprehension difficulties show poorer metacognitive monitoring on exactly this measure	[29]
Hints recorded as before or after a first attempt	Hints requested before any attempt are a stronger struggle signal than raw hint counts	[30]
Repair tracked (reread or revision after a hint)	Comprehension monitoring is detecting a difficulty *and* attempting a repair	[28]
Answer changes not treated as struggle on their own	Most answer changes on tests are from incorrect to correct (shown in adult test-takers)	[31]
Overconfidence on errors as a key signal	Poor comprehenders read fluently but overestimate their understanding, so teachers miss them	[25], [26]

3.3. Signals and Data Model

The application stores one `StudentResponse` per completed passage, containing one `QuestionResponse` per question. Each `QuestionResponse` records the selected answer, correctness, pre-answer confidence (1–4), postdictive judgement, hint count, whether a hint preceded the first attempt, answer changes, first-interaction delay, question time, passage revisits and whether a repair followed a hint. From these, the engine computes 17 session metrics, the most important of which are listed in Table 3.3.

Table 3.3: Key session metrics.

Metric	Definition
Accuracy (overall, literal, inferential)	Share correct, overall and by question type
Inferential gap	Literal accuracy minus inferential accuracy
Bias score	Mean of (judged correctness − actual correctness); positive = overconfident
Overconfidence on errors	Share of wrong answers the learner judged "correct"
Pre-attempt hints	Hints requested before any answer was selected
Processing signals	First-interaction delay, question time, revisits, answer changes

The same metric definitions are implemented in TypeScript (`recommendationEngine.ts`) and in the Python evaluation (`session_features()`), so the simulated evaluation measures the logic the app actually runs.

3.4. Recommendation Rules

Each rule produces an indicator with a severity, the evidence that triggered it (counts, not interpretation) and a plain-language interpretation for the facilitator. A learner's status is their highest severity: *high* is shown as "Needs support", *medium* as "Worth a check-in" and *low* as context only. Figure 3.2 shows a learner card from the dashboard.

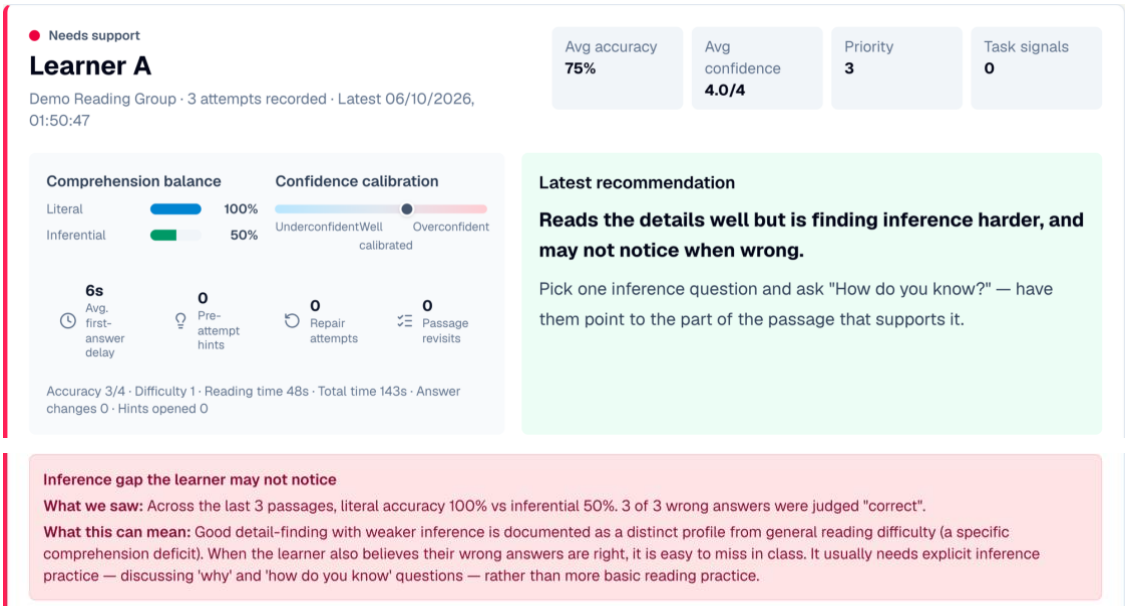


Figure 3.2: Facilitator dashboard learner card for demo Learner A (screenshot of the prototype, cropped): (top) status, comprehension balance, confidence calibration and the latest recommendation; (bottom) the high-severity indicator with its evidence and interpretation.

Rules v1 were the first version. Any single processing signal (a slow start, two revisits, two answer changes, one pre-attempt hint) raised a concern, and any incorrect answer raised at least a medium indicator.

Rules v2, the rule engine evaluated in the rest of this report, refined v1 using the research above:

- Processing difficulty requires two converging signals (two pre-attempt hints count as two), because single signals are common in typical readers.
- Accuracy below 50% is high, 50–74% medium, and a single miss is low.
- A new comprehension pattern fires when literal accuracy exceeds inferential accuracy by at least 0.5 *and* at least half of the wrong answers were judged "correct". Only this combination is high. A gap without overconfidence is low.
- Miscalibration alone stays medium, since some typical readers are also overconfident.
- When three passages are available, the comprehension pattern is computed over all 12 questions, with the gap threshold lowered to 0.34.

The dashboard also generates class-pattern alerts when a number of learners miss the same question or exhibit the same gap (again this is more likely to be a passage issue than an individual learner issue). All the thresholds are prototype defaults rather than validated cut-offs; they are listed in full in Appendix A.2.

3.5. Evaluation Method: Simulated Classrooms

No real classroom outcome data was available within the project's ethics approval and timeline, so there was no ground truth against which to measure the rules. The evaluation therefore used simulated learners, following the AIED tradition reviewed in Section 2.8. Because each simulated learner's profile is known, recall, precision and false-alarm rates can be computed directly. That is not possible with a handful of manually worked examples.

Five reader profiles (personas) were defined following the reader-profile literature (Table 3.4). Three are treated as at risk: 22% of learners, which came to about 6.7 per class of 30 on average in the simulated classrooms.

Table 3.4: Simulated reader profiles (full parameters in Appendix A.1).

Persona	Share	P(correct) literal / inferential	Behaviour	At risk
Typical	70%	0.90 / 0.80	Well calibrated, few hints	No
Careful reviser	8%	0.90 / 0.82	Slower, more revisits and answer changes, frequent repair	No
Poor comprehender	11%	0.85 / 0.45	Fluent and fast; judges 65% of wrong answers "correct"	Yes
Decoding-limited	6%	0.55 / 0.45	Slow, many pre-attempt hints, aware of difficulty	Yes
Mixed	5%	0.50 / 0.35	Slow, moderately overconfident	Yes

Only the 11% poor-comprehender share comes from data (24 of 218 children [26]). All other shares and distributions are assumptions shaped by the qualitative descriptions in the literature. Each simulated session generates four questions with the same fields as the app. Pre-answer confidence, judgements, hints, timings, revisits and answer changes are drawn from each persona's distributions.

Models were trained on 4,000 sessions (seed 101) and tested on 200 independent classrooms of 30 learners (6,000 sessions, seed 202). The rules were designed on neither set, but v2 was adjusted once after a first run (the overconfidence-on-errors measure was added) before being tested on the held-out seed. This is reported as a limitation.

3.6. Compared Systems

Seven systems were run on every test classroom:

1. **Score only:** flag if accuracy is below 75%. This is the baseline a facilitator could compute without the tool.
2. **Rules v1, "needs support" only:** flag if severity is high.
3. **Rules v1, "needs support" or "check-in":** flag if severity is medium or high.
4. **Rules v2, "needs support" only:** flag if severity is high.
5. **Rules v2, "needs support" or "check-in":** flag if severity is medium or high.
6. **Random Forest (trained on v1 rule labels):** the planned ML approach. With no real labels, the rules act as pseudo-labellers.
7. **Random Forest (oracle, "answer-key model"):** trained on the true simulated labels. This is a ceiling showing what a model could learn *if* real outcome labels existed. It cannot be deployed.

The Random Forests (scikit-learn, 200 trees, maximum depth 6) use all 17 session metrics. Random Forest was chosen over a black-box model because feature importances keep its predictions partly traceable to the input signals.

3.7. Metrics and Robustness Checks

For each system, the evaluation reports recall (share of at-risk learners flagged), precision (proportion of flags that are at-risk), false-alarm rate (proportion of not-at-risk learners flagged), and the proportion of poor comprehenders flagged. Each figure has a 95% bootstrap confidence interval from 1,000 resamples of the 200 test classrooms, and the headline runs were repeated on five independent train/test seed pairs.
Because the personas are assumptions, five checks test how much the results depend on them:

- **Timing-neutral (headline scenario):** poor comprehenders are given typical timing, hints, revisits, answer changes and repair. The only thing that differentiates them is their comprehension and self-judgement. This removes the rather unrealistically easy speed cue, so all of the headline results in Section 4 are based on this scenario. The results for the original personas ("as defined") are given for comparison.
- **Persona difference (α = 1, 0.75, 0.5):** each persona parameter is blended towards the typical persona, resulting in the groups being less distinct.
- **Poor-comprehender share:** 5%, 11% and 15%.
- **Class size:** 20 instead of 30.
- **More data per learner:** three passages pooled (12 questions) rather than one passage.

3.8. Implementation and System Features

The proof-of-concept tool is a Next.js 16 and TypeScript web application styled with Tailwind CSS. It provides a student activity flow and a facilitator dashboard with class and student

management. Data is stored in browser local storage, which keeps the proof of concept self-contained, and can be exported as CSV in the same column format the evaluation produces. The evaluation is a Python script (NumPy, pandas, scikit-learn) with pinned dependency versions. It runs end to end in about six minutes and regenerates every table and figure; Appendix B describes how to reproduce the results.

Student Activity. Design target is 5-10 minutes per check so it can run every session. The learner goes through a fixed sequence with a visible progress bar (in line with R2: welcome, instructions, story, 4 questions, reflection, finish). Reading time from the story screen is recorded. On each of the questions the learner selects an answer and rates their confidence from 1 to 4. They can ask for a hint or go back to the passage at any time. Before proceeding they must answer "Do you think you got that right?" (Yes, Not sure, No). The system records every answer change, every passage revisit, time to first interaction, time on the question and whether a hint was requested prior to any answer being selected. A repair is recorded if a hint is followed by a reread or a revision. The reflection asks for difficulty (1-3) and an optional comment. The interface is simple with large type, one question per screen and plain instructions to keep cognitive load low for the younger readers.

Facilitator Dashboard. The dashboard presents attempts in the form of a support profile for each learner. Each profile has a colour-coded status (“Needs support”, “Worth a check-in” or “On track”), a one-sentence summary and a suggested next step. Each indicator is listed with its evidence and interpretation, along with the learner’s attempt history. Profiles are listed by most recent attempt, and the summary counts at the top show how many learners have a 'needs support' attempt (R1). Where a learner has completed three passages, the comprehension-pattern check will pool all 12 questions, as one passage only provides two of each question type.

A class view filters by class and displays class-pattern alerts (R6). An alert is triggered when at least three learners (and at least half of those who attempted the activity) miss the same question or exhibit the same inferential gap. Classes and students can be added by the facilitator and all attempts can be exported as CSV for further analysis. There is a demo mode which loads seven anonymous learners whose sessions follow the evaluation personas. The same engine generates the recommendations for these learners so the screenshots such as Figure 3.2 show genuine rule output.

Keeping the App and the Evaluation Aligned. The rules are defined in two places (in TypeScript for the app and in Python for the evaluation). To ensure they are the same, the metric definitions and thresholds have the same names and values in both. The TypeScript source indicates which Python function each threshold maps to and the evaluation's per-learner CSV uses the column names as exported from the app. Formal automated parity tests between the two implementations have not been written and are flagged as future work.

4. Results

All results are on simulated learners. They show how the design behaves under the stated assumptions, not how it would perform in a classroom. Unless stated otherwise, figures show the timing-neutral scenario (Section 3.7) and one passage (four questions), the unit the app flags. They are averaged over 200 test classrooms with 95% bootstrap intervals. Results for the original “as defined” personas are given for comparison.

4.1. Overall Comparison (RQ1)

Table 4.1: Detection of at-risk learners, timing-neutral, one passage (n = 6,000 simulated learners; 6.7 at risk per class of 30 on average).

System	Recall (95% CI)	Precision	False-alarm rate	Flagged per class of 30	Poor comprehenders flagged
Score only (< 75%)	59.3% (56.2–62.2)	60.0%	10.8%	6.6	46.4%
Rules v1, needs support	48.8% (45.6–51.9)	68.5%	5.9%	4.7	25.3%
Rules v1, needs support or check-in	97.2% (96.1–98.2)	24.2%	87.1%	26.8	94.4%
Rules v2, needs support	**64.0% (61.0–66.9)**	**66.9%**	**8.8%**	**6.4**	**53.7%**
Rules v2, needs support or check-in	85.7% (83.6–87.7)	45.8%	28.9%	12.5	72.1%
Random Forest (v1 rule labels)	48.8% (45.6–51.9)	68.5%	5.9%	4.7	25.3%
Random Forest (answer-key, ceiling)	62.6% (59.4–65.8)	90.4%	1.9%	4.7	29.7%

At the “needs support” level, rules v2 caught 64.0% of at-risk learners, compared with 59.3% for the score alone. They did so with fewer false alarms (8.8% vs 10.8%) and higher precision (66.9% vs 60.0%). The confidence intervals overlap slightly, but the advantage held in all ten replication runs (five seed pairs, timing-neutral and as defined), with gains of 2.8 to 6.8 percentage points. With the original personas, the pattern is the same: 60.9% vs 56.7% at an equal false-alarm rate (10.2% vs 10.1%).

The v1 to v2 revision was most significant at the check-in level. v1 rules flagged 26.8 of the 30 learners as "worth a check-in" which is of no use to a facilitator. v2 rules flagged 12.5 learners (85.7% of at-risk learners), so the two tiers now work as they should: a short "needs support" list and a broader check-in list.

4.2. Detecting Poor Comprehenders (RQ2)

Table 4.2: Share of each reader profile flagged, timing-neutral, one passage.

System	Typical	Careful reviser	Poor comprehender	Decoding-limited	Mixed
Score only	10.8%	10.5%	46.4%	68.5%	82.2%
Rules v2, needs support	8.4%	11.7%	53.7%	70.5%	83.2%
Random Forest (answer-key)	1.7%	3.4%	29.7%	100%	95.5%

Poor comprehenders were the group that was most difficult for every system as the literature predicts. Rules v2 flagged 53.7% of them compared with 46.4% for the score, and flagged fewer of the typical readers (8.4% vs 10.8%). It did flag slightly more careful revisers (11.7% vs 10.5%), likely because their extra revisits and answer changes sometimes meet the processing rule. The comprehension-pattern rule is the explanation for the difference: it picks up the learners whose overall score is in fact acceptable but who are getting their inferential answers wrong and wrong with confidence. The advantage is held in all five of the timing-neutral seed pairs (52.0–55.4% vs 44.3–47.6%).

Comparison with the Original Personas. As first defined, poor comprehenders also answer faster than typical readers. In that run, rules v2 flagged 50.1% of them against 42.2% for the score, and the answer-key model flagged 57.7%. Removing the speed cue cut the answer-key model to 29.7%, below the score, which shows it had relied mainly on speed. The rules barely changed (50.1% → 53.7%), because they never used speed to detect this group (Figure 4.1).

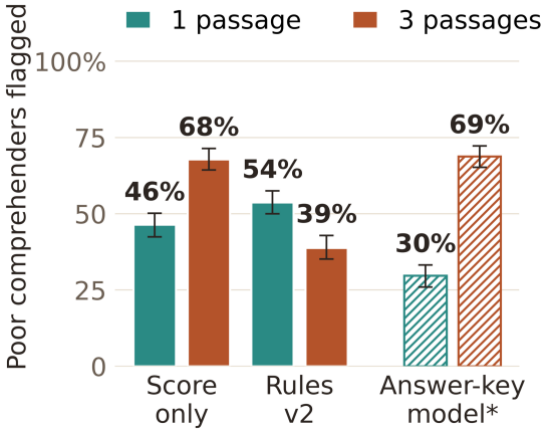


*Figure 4.1: Poor comprehenders flagged, timing-neutral, one vs three passages, with 95% confidence intervals. The answer-key model (marked *) is trained on the true simulated labels and is a ceiling, not a deployable tool.*

4.3. More Data per Learner

Because the check is designed to run every session, three passages correspond to about three sessions of checks. Pooling them (12 questions) improved most systems (Figure 4.2, Table 4.3).

Table 4.3: One vs three passages, timing-neutral.

System	Recall, 1 → 3 passages	False-alarm rate, 1 → 3	Poor comprehenders, 1 → 3
Score only	59.3% → 81.2%	10.8% → 8.8%	46.4% → 67.9%
Rules v2, needs support	64.0% → 66.5%	8.8% → 1.1%	53.7% → 38.8%
Rules v2, needs support or check-in	85.7% → 87.9%	28.9% → 12.4%	72.1% → 76.4%
Random Forest (answer-key)	62.6% → 84.2%	1.9% → 1.2%	29.7% → 68.8%

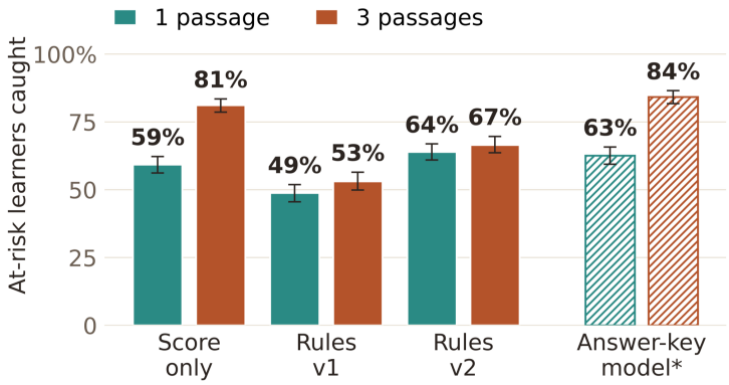


Figure 4.2: Recall of all at-risk learners, one vs three passages, timing-neutral, with 95% confidence intervals.

With three passages, rules v2 became very precise: 94.3% of "needs support" flags were correct, compared with 72.5% for the score, and only 1.1% of not-at-risk learners were flagged.

However, the rules flagged *fewer* poor comprehenders than with one passage (38.8% vs 53.7%), and fewer than the score (67.9%). The score and the answer-key model both caught more at-risk learners overall (81.2% and 84.2% vs 66.5%).

The cause is a threshold-granularity issue found after the results were produced. Over six inferential questions, the gap moves in steps of 1/6. A gap of 2/6 (0.333) falls just below the pre-specified 0.34 threshold, so the pooled rule behaves like the single-passage threshold of 0.5. A post-hoc run with 0.33 raised poor-comprehender detection to 60.0% and recall to 77.1% at a 7.3% false-alarm rate. This is reported separately because it was not the pre-specified rule.

4.4. Machine Learning Comparison (RQ3)

The Random Forest trained on the rule-generated labels did not outperform the rules at all. In the timing-neutral run the flags were exactly the same as rules v1 (48.8% recall each). With the original personas they were virtually identical (45.7% vs 46.0%). Without any independent labels a model can only learn the logic that was used to generate its labels. The feature importances (as defined run) bear this out with accuracy (0.38) and inferential accuracy (0.15) dominating, which mirrors the rule conditions.

The answer-key model shows what real labels could be worth, with a caveat. In the timing-neutral run, it reached 62.6% recall at a 1.9% false-alarm rate on one passage, level with the rules. On three passages, it reached 84.2%, clearly above them. With the original personas, it reached 77.1% and 97.4%, but its top features there were total time (0.31), question time (0.22) and first-interaction delay (0.14). These are the timing parameters assumed for each persona, so much of its apparent advantage reflects the simulation's assumptions rather than validated signal value.

4.5. Robustness and Sensitivity

Table 4.4: Recall under alternative assumptions, one passage. Each scenario varies the original personas; the timing-neutral row is the headline scenario used elsewhere in this section.

Scenario	Score only	Rules v2, needs support
As defined (α = 1, 11% poor comprehenders, 30 learners)	56.7%	60.9%
Personas less distinct, α = 0.75	46.4%	49.8%
Personas less distinct, α = 0.5	31.3%	34.0%
Poor-comprehender share 5%	64.1%	68.7%
Poor-comprehender share 15%	55.4%	61.7%
Class size 20	58.4%	63.6%
Timing-neutral (headline)	59.3%	64.0%

Rules v2 remained 2.7 to 6.3 points ahead in every scenario so the ordering is stable. The actual performance is of course very much dependent on how well the personas are separated. Halving the difference between the profiles (α=0.5) resulted in a near halving of recall for every system and the answer-key model dropped to 28.3%. The headline figures should therefore be treated as relative rather than absolute.

4.6. Workload

An average class of 30 contains about 6.7 at-risk learners in the simulation. Rules v2 put 6.4 learners per class on the "needs support" list and 12.5 on the wider check-in list (Figure 4.3). Each flag comes with the evidence and a suggested next step, rather than the raw interaction data behind it.

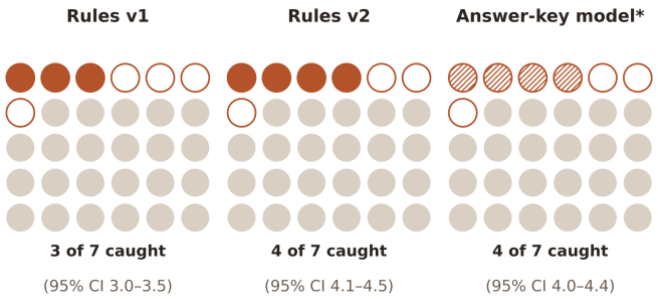


Figure 4.3: At-risk learners caught in a class of 30 (about 7 at risk), timing-neutral, one passage. Filled: caught; outlined: missed; grey: not at risk.

4.7. End-to-End Traceability Walkthrough

The metrics above show how often the rules are right. This walkthrough shows *how* a single recommendation is produced and what the facilitator sees, using demo Learner A, a poor-comprehender profile, run through the app's actual engine.

1. Raw Interactions. On each of three passages, Learner A answers both literal questions and one inferential question correctly, and misses the other inferential question. They judge all four answers "correct", with confidence 4. They start each question within about 6 seconds, spend 16–22 seconds per question, use no hints, do not revisit the passage and rate the activity "easy".

2. Session Metrics. For the most recent passage only: accuracy 75%, literal 100%, inferential 50%, bias score +0.25, no pre-attempt hints and no processing signals. Pooled across the three passages: literal 6/6, inferential 3/6, an inferential gap of 0.50 and 3 of 3 wrong answers judged correct (overconfidence on errors of 1.0).

3. Rules. No processing rule fires as the learner is quick and does not seek help. The accuracy rule fires at *low* (one miss). The bias score is below 0.4 so the miscalibration rule does not fire. The comprehension-pattern rule fires at *high as both* the pooled gap ($0.50 \geq 0.34$) and the overconfidence on errors ($1.0 \geq 0.5$) are satisfied.

4. Facilitator View. Status: *Needs support*. Summary: "Reads the details well but is finding inference harder, and may not notice when wrong." Indicator: "Inference gap the learner may not notice", with the evidence "Across the last 3 passages, literal accuracy 100% vs inferential 50%. 3 of 3 wrong answers were judged 'correct'." Suggested next step: pick one inference question and ask "How do you know?", having the learner point to the supporting part of the passage (Figure 3.2).

5. Comparison. A score threshold would not flag Learner A at all, since 75% is not below 75%. Rules v1 would show only "Worth a check-in", on a check-in list of 26.8 learners per class. A facilitator relying on observation would see a quick, confident, fluent reader. Only the combination of the inferential gap and the learner's own judgements reveals the pattern, and every number in the recommendation can be traced back to the four answers on each passage.

The example is constructed to show the pattern clearly. It illustrates traceability, not typical accuracy: Section 4.2 shows that nearly half of simulated poor comprehenders are still missed.

5. Discussion

5.1. Answering the Research Questions

RQ1: Rules vs Score. The evidence-linked rules performed only slightly better than a score threshold. Recall was 4.7 percentage points higher (64.0% vs 59.3%) with fewer false alarms (8.8% vs 10.8%), and the advantage was consistent across all ten replications and all sensitivity scenarios. The gain is very small, and a facilitator could not tell the two systems apart on a single class. The more important result is qualitative: the rules give the same or better detection, *and they explain why.* For example, a score of 2 out of 4 indicates that the learner has struggled, whereas an indicator such as "literal 100%, inferential 0%, both wrong answers judged correct" provides actionable guidance for subsequent steps.

RQ2: Poor Comprehenders. On one passage the comprehension-pattern rule flagged more poor comprehenders than the score (53.7% vs 46.4% and 50.1% vs 42.2% with the original personas). This again supports the design choice of pairing the inferential gap with overconfidence on errors. A rather significant proportion of the poor comprehenders (nearly half) still went unflagged. With three passages and the pre-specified threshold the score clearly outperformed the rule (67.9% vs 38.8%). The rule is promising but not yet reliable and the pooled threshold will need adjusting (Section 6.2).

RQ3: Machine Learning. With no real labels the Random Forest added nothing. It simply replicated the rules that had been used to label the training data in the timing-neutral run. This is the most transferable result of the project. In a cold-start scenario an ML model trained on rule-generated pseudo-labels will inherit the rules' blind spots and so cannot be presented as an improvement. The answer-key model matched the rules on one passage (62.6%) and beat them on three (84.2%) which again suggests what real labels could do once more data per learner is available. Its superior performance with the original personas (77.1% and 97.4%) was largely due to the speed cue that the simulation itself introduced.

5.2. Relation to Literature

The results are in line with the gaps identified in Section 2.6. The literature states that AI in K–12 should be a complement to the teacher rather than a replacement [5] and that younger learners require more structure than autonomy (Section 2.4). The prototype is designed to embody this by keeping the learner's task very simple and structured and having the analytics report back to the facilitator rather than the child.

The dependence-on-teacher-expertise gap (Section 2.6) is dealt with head on. Schmitterer and Brod [27] found that teachers' judgements of reading difficulty track spelling much more closely than comprehension. The poor comprehender profile is precisely the case in which an observer relying on surface fluency is going to be misled. A tool which exposes the inference gap and the learner's overconfidence is exactly the kind of structured support that the literature suggests less experienced facilitators are lacking.

Huang et al. [9] found that the educational value of AI is very much dependent on the quality of implementation. Our results at this small scale are consistent with that: the same signals produced a near-useless check-in list under v1 (26.8 of 30 learners) and a workable one under

v2 (12.5). It is as much to do with how the signals are combined as it is with which signals are collected.

Ultimately the evaluation methodology is the classic simulated learner approach [35], [37] and inherits its key flaw. Käser and Alexandron [40] maintain that simulated learners have to be compared to real learners via a 'Turing-like' test before their results can be considered valid. The personas in this project have not undergone such a test.

5.3. Contribution

The project contributes:

1. A working facilitator decision-support prototype in which every flag is linked to the evidence underpinning it and to a plain-language explanation.
2. An activity and feature design which takes specific reading comprehension findings and turns them into tangible signals: postdictive judgement, pre-attempt hints, literal vs inferential accuracy and overconfidence on errors.
3. A reproducible simulated-classroom evaluation with confidence intervals, seed replication and sensitivity analysis, which other prototype studies can reuse before real data is available.
4. Evidence that pseudo-label ML adds no value over its labelling rules in a cold-start setting, which cautions projects that bolt ML onto rule-based systems.

5.4. Design Trade-offs and Ethical Considerations

Missing a Learner vs Wasting Attention. No threshold maximises both recall and precision, so the dashboard has two tiers. The “needs support” tier is slightly more selective than a score threshold (8.8% vs 10.8% false alarms) and suits a facilitator with little time. The check-in tier catches 86% of at-risk learners but also flags 29% of the rest. Where a missed learner might get no other help, the wider tier is the safer default. Which tier a programme leads with depends on class size and facilitator time, and the system should make that choice visible rather than hide it in a threshold.

Transparency vs Accuracy. With the original personas, the answer-key model was far more accurate than the rules, but its strongest features were the timing parameters built into the simulation, and on one passage its advantage disappeared once that cue was removed. An opaque model can learn an artefact of its training data and present it with confidence. Because the rules are inspectable, this project found its own error, the pooled threshold, just by reading the logic. Once real labels exist, a hybrid may work best: a model ranks learners, and rules supply evidence the facilitator can check.

Labelling Children. A false positive costs a facilitator a conversation. It could also attach a label to a child, so the wording describes patterns in an activity rather than traits of a learner (“may need”, “worth a check-in”, “may not notice when wrong”). The check is ungraded, learners or their families should be able to opt out, and the status is shown only to the facilitator. The “On track” text states that it “reflects the current rules’ coverage, not a guarantee”.

Over-reliance on the Tool. Decision aids can produce automation bias, in which users accept a recommendation without checking it [41]. That risk is highest for exactly the less-experienced facilitators this tool targets. Showing evidence alongside each flag is a partial safeguard, since

it invites the facilitator to check. Whether facilitators actually do so is an open question for the facilitator study in Section 6.2.

Equity and Bias. An inferential gap can come from vocabulary or background knowledge rather than comprehension, for example for English-language learners or when a passage assumes unfamiliar cultural knowledge. The comprehension-pattern rule could then mistake a language or content gap for a reading difficulty. Fairness therefore depends on passage choice and on validating the tool across different learner groups, not just on accuracy.

Privacy. The data concern children. The prototype keeps all data in the facilitator's browser and collects no identifiers beyond the name a facilitator enters. Any real deployment would need informed consent, secure storage and compliance with the New Zealand Privacy Act 2020 [42].

5.5. Limitations and Threats to Validity

- **Assumed Personas.** Only the 11% poor-comprehender share comes from data. All other parameters are assumptions, and the α runs show that absolute performance depends heavily on them. The personas are also generated from the same signals the rules read, which favours the rules. Results are therefore relative comparisons under stated assumptions, not estimates of classroom accuracy.
- **Rules Tuned Once on Simulated Data.** Rules v2 were adjusted after a first run, then tested on a held-out seed. A held-out seed drawn from the same personas cannot rule out overfitting to those personas.
- **Pooled Threshold Error.** The 0.34 threshold behaves like 0.5 over three passages (Section 4.3). The pre-specified result is reported; the corrected threshold is post hoc.
- **Narrow Task.** The evaluation covers one activity type: three short narrative passages, multiple-choice questions and four questions per passage. Generalisation to other subjects, ages or question formats is untested.
- **No Human Evaluation.** No facilitator used the dashboard, so usability, trust and whether recommendations change facilitator actions are unknown. The workload figures in Section 4.6 support the workload-reduction aim but do not demonstrate it.
- **Prototype Infrastructure.** Browser local storage is not suitable for real learner data, which would need authentication, server-side storage and privacy safeguards.
- **One Half of a Split Project.** Lesson design, delivery and AI-supported content generation belong to StudyBuddy and are evaluated in the partner report. This report evaluates the decision-support tool alone, and the two prototypes have not been integrated or tested together.

6. Conclusions and Future Work

6.1. Conclusions

This report was designed to see if interaction data from a brief reading task could be used to generate evidence-based recommendations to help facilitators spot Year 3–6 learners who require support. A proof-of-concept decision-support tool was developed, and its rule engine was tested on 6,000 simulated learners. In the conservative timing-neutral scenario, the rule engine modestly exceeded the accuracy score on a single passage (64.0% vs 59.3%) with fewer false positives. It also identified more of the poor comprehenders that teachers are likely to

overlook (53.7% vs 46.4%) with the evidence for each flag. With three passages pooled, however, it flagged fewer poor comprehenders than the score (38.8% vs 67.9%) because of the pooled-threshold issue. A Random Forest trained without any real labels added nothing beyond the rules it learned. A model trained on the true labels performed on par with the rules on one passage and exceeded them on three, which indicates what real outcome data would be able to do.

The answer to the research question is therefore a qualified yes. The approach is technically feasible and is in fact somewhat better than a score under the stated assumptions. It is a tool to assist the facilitator in identifying those learners who do need support rather than a tool to replace their judgement. For facilitator-based education the project demonstrates that a lightweight, transparent tool can carry some of the diagnostic expertise that would otherwise be required of every facilitator in the model of separating lesson design from delivery.

6.2. Future Work

In order of priority:

1. **Validate with Real Learners.** Gather consented, anonymised session data from Year 3–6 classes along with an independent reading assessment. Use this to check whether the personas are in fact akin to real learners (in the spirit of Käser and Alexandron's Turing-like test [40]) and to re-calibrate the rule thresholds.
2. **Fix the Pooled Gap Threshold.** Set it to a value consistent with the 1/6 step (e.g. 0.33) and assess it as a new pre-specified rule on new data.
3. **Use Repeated Checks Over Time.** Include before and after comparisons, trend views and a per-learner baseline and over the longer term use those insights to personalise the learning.
4. **Facilitator Study.** Observe facilitators using the dashboard to test whether the evidence and interpretations are understood, trusted and acted on, and whether the tool reduces their workload.
5. **Train ML on Real Labels.** Once outcome data exists, retrain the Random Forest on it and compare against the rules. Use timing features only if they prove predictive in real data.
6. **Broaden the Activity.** Add more passages, genres and question formats, and other subjects. Modular design allows for flexibility with future student activities and data.
7. **Integrate with StudyBuddy.** Add the reading check as a StudyBuddy lesson block, store results in StudyBuddy's authenticated database with privacy safeguards for children's data, and show the support view during delivery (Section 1.3).

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Appendix A Simulation Parameters and Rule Thresholds

A.1. Persona Parameters

All values below are assumptions shaped by the reader-profile literature, except the 11% poor-comprehender share, which comes from Kelso et al. Probability triples give P(judged correct, unsure, incorrect) or P(difficulty 1, 2, 3). Timings are mean (standard deviation) in seconds. In the timing-neutral scenario, the poor comprehender's timing, hint, revisit, answer-change and repair parameters are set equal to the typical persona's.

Table A.1: Persona parameters used in the simulation (`simulated_classroom_evaluation_v2.py`).

Parameter	Typical	Careful reviser	Poor comprehender	Decoding-limited	Mixed
Share of class	70%	8%	11%	6%	5%
P(correct), literal	0.90	0.90	0.85	0.55	0.50
P(correct), inferential	0.80	0.82	0.45	0.45	0.35
Judgement if correct	.75/.20/.05	.78/.18/.04	.90/.08/.02	.50/.35/.15	.60/.30/.10
Judgement if wrong	.25/.35/.40	.20/.30/.50	.65/.20/.15	.15/.30/.55	.45/.30/.25
Mean confidence (1–4)	3.2	3.0	3.4	2.2	2.6
P(hint before attempt)	0.05	0.05	0.03	0.35	0.25
P(hint after attempt)	0.15	0.25	0.08	0.35	0.30
First-interaction delay (s)	9 (4)	12 (5)	7 (3)	24 (9)	17 (7)
Question time (s)	30 (10)	40 (12)	22 (8)	70 (20)	55 (18)
Answer changes per question	0.2	0.7	0.15	1.0	0.9
Passage revisits per question	0.2	0.5	0.15	1.5	1.2
P(repair after hint)	0.5	0.8	0.3	0.6	0.5
Perceived difficulty	.60/.35/.05	.45/.45/.10	.65/.30/.05	.10/.40/.50	.20/.40/.40
Reading time (s)	60 (15)	70 (15)	50 (12)	120 (30)	100 (25)
At risk	No	No	Yes	Yes	Yes

Pre-answer confidence is drawn from a normal distribution with the persona mean, shifted by +0.3 for a correct answer and −0.3 for a wrong one (standard deviation 0.7), then rounded and clipped to 1–4. The probability of a hint after an attempt is multiplied by 1.5 when the answer is wrong. Answer changes and revisits are Poisson counts.

A.2. Rule Thresholds

Table A.2: Indicator conditions in rules v1 and v2 (high = "Needs support", medium = "Worth a check-in", low = context only).

Indicator	Rules v1	Rules v2
Processing difficulty	Any one of: first delay ≥ 20 s, answer changes ≥ 2, revisits ≥ 2, question time ≥ 75 s, pre-attempt hints ≥ 1. Medium if accuracy ≥ 75%, else high	Two or more of: first delay ≥ 20 s, revisits ≥ 2, question time ≥ 75 s, answer changes ≥ 3, pre-attempt hints ≥ 1, pre-attempt hints ≥ 2. Medium if accuracy ≥ 75%, else high
Accuracy	Any miss: medium; below 50%: high	Below 50%: high; 50–74%: medium; 75–99%: low
Hint use	One hint: low; more than one: medium	Low note only
Low confidence	Any answer rated 1–2: medium	Two or more answers rated 1–2: medium

Indicator	Rules v1	Rules v2
Perceived difficulty	Rated 3: medium	Rated 3 and accuracy below 75%: medium
Miscalibration	Bias ≥ 0.4: medium; ≥ 0.7: high	Bias ≥ 0.4: medium
Inferential gap / comprehension pattern	Gap ≥ 0.34: medium	Gap ≥ 0.5 (one passage) or ≥ 0.34 (three pooled): high if overconfidence on errors ≥ 0.5, otherwise low
Extended time	Total time > 240 s: medium	Low note only

Appendix B Reproducing the Results

All code is in the project repository (https://github.com/mtha724/partIV-facilitator-decision-support). The evaluation lives in `facilitator-decision-support/evaluation/` and the application in `facilitator-decision-support/src/`.

Evaluation. Python 3.9 or later with pinned versions (NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, Matplotlib 3.9.4):

```
pip install -r requirements.txt
python3 simulated_classroom_evaluation_v2.py
python3 make_poster_figures.py
```

The first script takes about six minutes and writes every table to `results_v2/`; `--quick` skips the seed replication. The second produces the figures in `figures/`. Models train on seed 101 (4,000 sessions) and are tested on seed 202 (200 classrooms of 30). Seed replication repeats the main 101/202 pair with four more: 10101/10202, 20101/20202, 30101/30202 and 40101/40202. Bootstrap intervals use seed 7 with 1,000 resamples of the test classrooms.

Application. Node.js with Next.js 16: run `npm install` and then `npm run dev` in `facilitator-decision-support/`, and open the student activity or the facilitator dashboard. The dashboard's demo mode loads the seven demo learners used for Figure 3.2.